

1. A telecom company notices that 30% of customers who made complaints in the last three months have left. However, 70% of customers who didn't complain remained. How would you modify your feature engineering strategy to improve churn prediction?

Model type - Classification

Step 1: Collect the dataset using read_csv function

Step 2: Data preprocessing- remove null values encode categorical data

Step 3: choose independent & dependent variables

Step 4: split train & test data set respectively

Step 5: Build classification model using logistic regression, or SVM or random forest

Step 6: Measure accuracy, precision, f1 to find out which model is best to predict churn rate

2. A warehouse manager observes that demand for certain products fluctuates seasonally, but your model is not capturing these patterns. How would you improve the model's predictive power?

Model type - Regression

Step 1: Collect the products dataset using read_csv function

Step 2: Data processing - handle missing data & encode categorical data

Step 3: Choose independent & dependent variables

Step 4: split train & test set respectively

Step 5: Build Regression model using linear regression,SVR, decision tree

Step 6: Measure accuracy using RSquare, SSE or SSR to figure out which model is best to determine the product demands

3. Your bank's fraud detection model has a high false positive rate, flagging many legitimate transactions as suspicious. What steps would you take to reduce false alarms while still detecting money laundering effectively?

Model type: Classification

Step 1: Collect the fraud detection dataset using read_csv function

Step 2: Data preprocessing - handle missing data & encode categorical data

Step 3: Choose independent & dependent variables

Step 4: split train & test set respectively

Step 5: Build classification model using random forest, naive bayes, KNN, decision tree

Step 6: measure accuracy, f1, classification matrix, precision to figure which model is best to detect money laundering effectively

4. An airline wants to use weather data in its flight delay prediction model, but 20% of past records lack weather conditions. How would you handle missing weather data without compromising model accuracy?

Model type - Regression

Step 1: Collect the flight schedule dataset using read_csv function

Step 2: Data processing - handle missing data using imputer & encode categorical data

Step 3: Choose independent & dependent variables

Step 4: split train & test set respectively

Step 5: Build Regression model using linear regression, SVR, decision tree

Step 6: Measure accuracy using RSquare, SSE or SSR to figure out which model is best to determine the flight delay

5. A student with high quiz scores but low engagement time is placed in an advanced learning path by your clustering model. However, they struggle with the content. How would you refine your clustering approach?

Model type - Clustering

Step 1: Collect the quiz dataset using read_csv function

Step 2: Data processing - handle missing data & encode categorical data

Step 3: Create a model using clustering methods like DBSCAN, OPTICS, agglomerative clustering, K means etc.

Step 4: Add Hypertuning to the model & using elbow method choose the best model

6. Your model predicts unusually low rental prices for high-end properties in luxury neighborhoods. What factors could be missing from the dataset, and how would you improve your model?

Model type - Regression

Step 1: Collect the rent prices dataset using read_csv function

Step 2: Data processing - handle missing data using imputer & encode categorical data

Step 3: Choose independent & dependent variables

Step 4: split train & test set respectively

Step 5: Build Regression model using linear regression,SVR, decision tree

Step 6: evaluate the model accuracy using RSquare, SSE or SSR to figure out which model is best to predict the rental price

7. Your anomaly detection system raises an alert about a network traffic pattern, but IT specialists confirm it's a routine software update. How would you improve the system to reduce such false positives?

Model type: Classification

Step 1: Collect the anomaly detection dataset using read_csv function

Step 2: Data preprocessing - handle missing data & encode categorical data

Step 3: Choose independent & dependent variables

Step 4: split train & test set respectively

Step 5: Build classification model using random forest, naive bayes, KNN, decision tree

Step 6: evaluate the model using accuracy, f1, classification matrix, precision to figure which model is best to detect anomaly in network traffic

8. A hospital's classification model struggles to differentiate between flu and COVID-19 because many symptoms overlap. What techniques can improve the model's ability to distinguish between these conditions?

Model type: Classification

Step 1: Collect the diseases dataset using read_csv function

Step 2: Data preprocessing - handle missing data & encode categorical data

Step 3: Choose independent & dependent variables

Step 4: split train & test set respectively

Step 5: Build classification model using random forest, naive bayes, KNN, decision tree

Step 6: evaluate the model using accuracy, f1, classification matrix, precision to figure which model is best to differentiate flu & COVID-19

9. Your ad engagement prediction model performs well on historical data but fails to generalize to new ad formats. What strategies can you use to improve its adaptability?

Model type - Regression

Step 1: Collect the ads dataset using read_csv function

Step 2: Data processing - handle missing data using imputer & encode categorical data

Step 3: Choose independent & dependent variables

Step 4: split train & test set respectively

Step 5: Build Regression model using linear regression,SVR, decision tree

Step 6: evaluate the model accuracy using RSquare, SSE or SSR to figure out which model is best to predict the ad engagement

10. A recruiter tries to bypass the fake job detection model by slightly modifying fraudulent job descriptions. How would you make the model more robust against such adversarial tactics?

Model type: Classification

Step 1: Collect the job description dataset using read.scv

Step 2: Data preprocessing - handle missing data & encode categorical data

Step 3: Choose independent & dependent variables

Step 4: split train & test set respectively

Step 5: Build classification model using random forest, naive bayes, KNN, decision tree

Step 6: evaluate the model using accuracy, f1, classification matrix, precision to figure which model is best to differentiate fake job & real job